

Mandatory Assignment 1

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Exercise 1

We download monthly data for the S&P 500 index ($\hat{\text{GSPC}}$) and the 13-week Treasury Bill rate ($\hat{\text{IRX}}$) from Yahoo! Finance for the period January 1, 2000 to February 28, 2026. The $\hat{\text{IRX}}$ series is an annualized yield in percent, so it is converted into a monthly risk-free rate as

$$r_{f,t} = (1 + \text{IRX}_t/100)^{1/12} - 1.$$

Monthly excess market returns are then computed as

$$\tilde{r}_{m,t} = r_{m,t} - r_{f,t},$$

where $r_{m,t}$ denotes the monthly return on the S&P 500 index. We use the estimated monthly mean excess return μ_m and monthly variance σ_m^2 to calibrate the synthetic asset universe used in the remainder of the assignment.

Table 1: Estimated monthly and annualized market parameters.

Parameter	Monthly	Annualised
Mean excess return	0.0045	0.0545
Variance	0.0019	0.0228

Note: Based on monthly S&P 500 excess returns ($\hat{\text{GSPC}}$) from January 2000 to February 2026. The risk-free rate is $\hat{\text{IRX}}$, converted to a monthly rate.

Table 1 reports the estimated monthly mean excess market return and variance, together with their annualized counterparts. The estimated monthly mean excess return is approximately 0.45%, while the annualized mean excess return is approximately 5.45%. The corresponding annualized variance is approximately 0.0228.

Exercise 2

Next, we construct a universe of $N = 50$ synthetic assets following the single-factor model

$$\tilde{r}_{i,t} = \beta_i \tilde{r}_{m,t} + \varepsilon_{i,t}.$$

The factor loadings β_i and idiosyncratic volatilities $\sigma_{\varepsilon,i}$ are drawn from the distributions specified in the assignment. Using the estimated market moments from Exercise 1, we construct the true mean vector μ and covariance matrix Σ , which are treated as the true parameters throughout the remainder of the assignment.

Table 2: Summary statistics for the synthetic asset universe.

Parameter	Min	Mean	Max
Beta	0.523	0.968	1.4876
Idiosyncratic volatility	0.029	0.0604	0.0865
Expected excess return	0.0024	0.0044	0.0068

Note: Summary statistics for the $N = 50$ synthetic assets.
All quantities are reported in monthly units.

Table 2 summarizes the characteristics of the $N = 50$ synthetic assets. The true covariance matrix is constructed as

$$\Sigma = \sigma_m^2 \beta \beta' + \text{diag}(\sigma_\varepsilon^2),$$

where the first term captures systematic market risk and the second term captures idiosyncratic risk.

Exercise 3

Using the true mean vector μ and covariance matrix Σ constructed in Exercise 2, we implement functions to compute the tangency portfolio and the mean–variance efficient frontier.

The efficient frontier consists of portfolios that minimize portfolio variance for a given target return,

$$\omega_{eff}(\bar{\mu}) = \arg \min_{\omega} \omega' \Sigma \omega,$$

subject to

$$\omega' \iota = 1, \quad \omega' \mu \geq \bar{\mu}.$$

The tangency portfolio is computed as

$$\omega_{tan} = \frac{\Sigma^{-1} \tilde{\mu}}{\iota' \Sigma^{-1} \tilde{\mu}},$$

where $\tilde{\mu} = \mu - r_f$ denotes expected excess returns.

The implemented functions return portfolio weights together with the corresponding expected return, variance, volatility and Sharpe ratio. They are used throughout the remainder of the assignment to evaluate portfolio allocations under both true and estimated parameters.

Exercise 4

Using the true parameter values μ and Σ , we compute the mean-variance efficient frontier and the corresponding tangency portfolio. The resulting frontier describes the best attainable risk-return combinations under the true data-generating process.

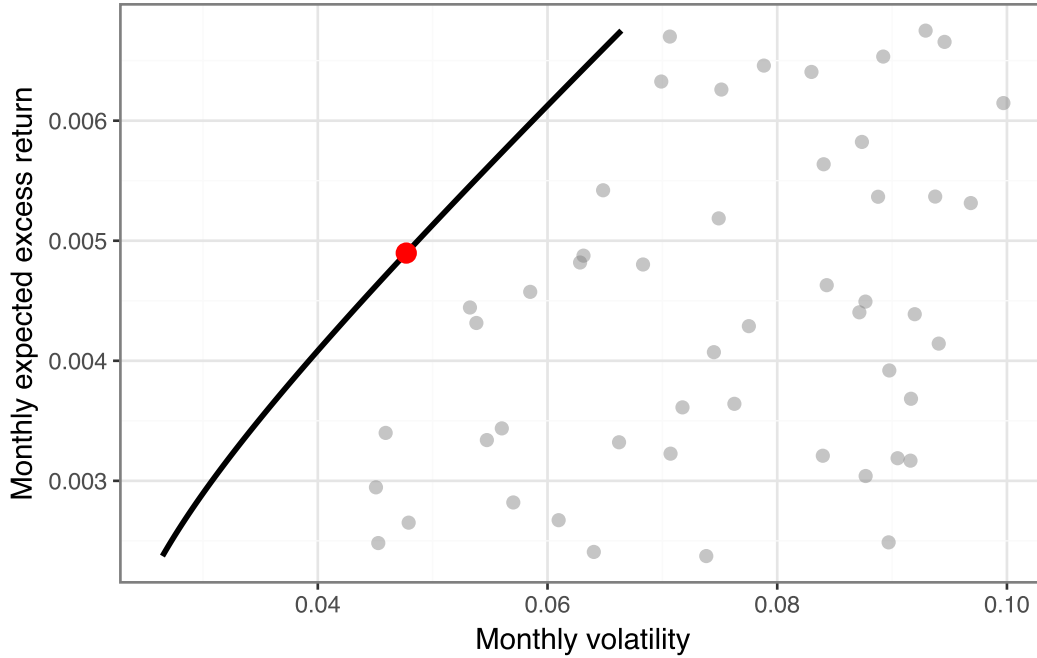


Figure 1: Mean-variance efficient frontier and tangency portfolio based on the true parameters.

Figure 1 shows the risk-return characteristics of the 50 synthetic assets together with the mean-variance efficient frontier. The grey points represent individual assets, while the frontier shows the set of portfolios that achieve the highest expected excess return for a given level of risk. The marked point denotes the tangency portfolio, which maximizes the Sharpe ratio and therefore represents the most attractive risk-return trade-off under the true parameter values.

Exercise 5

To study estimation error, we simulate asset returns from the true data-generating process and estimate the mean vector $\hat{\mu}$ and covariance matrix $\hat{\Sigma}$ from the simulated sample.

Using this function, we repeatedly simulate returns (15 times) and construct plug-in efficient frontiers based on the estimated parameters $\hat{\mu}$ and $\hat{\Sigma}$. We consider sample sizes of $T = 60$, $T = 120$

and $T = 500$ to examine how the magnitude of estimation error depends on the amount of available data.

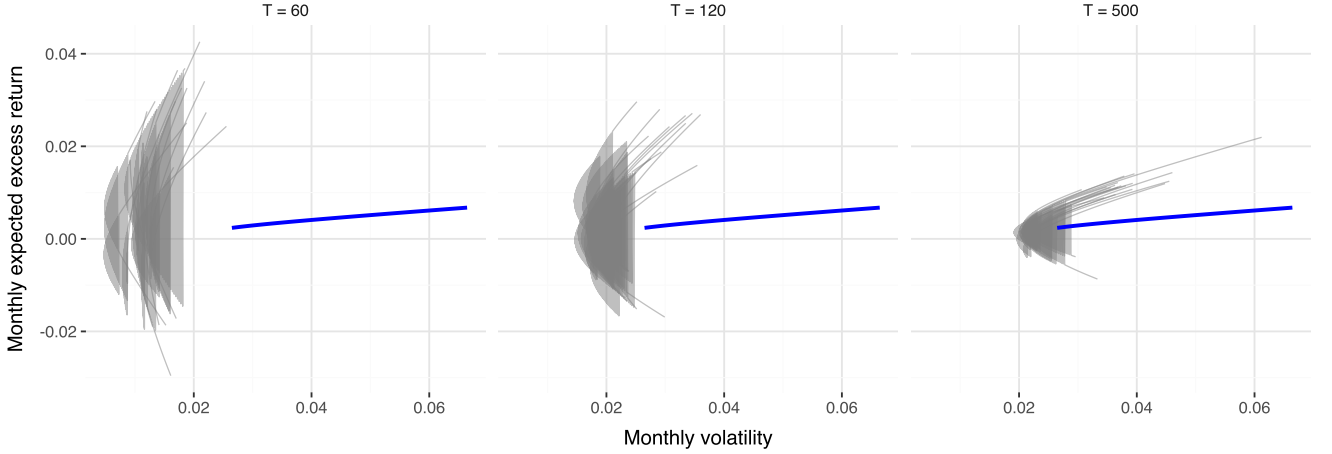


Figure 2: Estimated efficient frontiers from repeated simulations compared with the true frontier.

Figure 2 compares the true efficient frontier with frontiers estimated from simulated samples of different lengths. For $T = 60$, the estimated frontiers display substantial variation across simulations and often differ markedly from the true frontier. As the sample size increases to $T = 120$ and $T = 500$, the estimated frontiers become increasingly clustered around the true frontier. This indicates that larger samples produce more reliable estimates of the mean vector and covariance matrix, leading to more accurate estimates of the efficient frontier. Overall, the figure shows that estimation error becomes less severe as the amount of available data increases.

To evaluate the impact of estimation error, we compare the true tangency portfolio with the plug-in tangency portfolio. The true tangency portfolio is constructed using the true parameters μ and Σ , while the plug-in tangency portfolio is based on the estimated sample moments $\hat{\mu}$ and $\hat{\Sigma}$.

The tangency portfolio weights are given by

$$w_{tan} = \frac{\Sigma^{-1}\mu}{\iota^\top \Sigma^{-1}\mu},$$

where ι denotes a vector of ones.

The Sharpe ratio of a portfolio with weights w is defined as

$$SR = \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}}.$$

To analyze the effect of estimation error, asset returns are simulated from the multivariate normal distribution

$$r_t \sim \mathcal{N}(\mu, \Sigma).$$

For each simulated sample with size T , the sample estimates $\hat{\mu}$ and $\hat{\Sigma}$ are computed and used to construct the plug-in tangency portfolio. The resulting portfolio is then evaluated using the true parameters μ and Σ .

Table 3: Average Sharpe ratios of the true and plug-in tangency portfolios across sample sizes.

	T	true_sharpe	plugin_sharpe
0	60	0.102665	0.006393
1	120	0.102665	0.011123
2	240	0.102665	0.015163
3	480	0.102665	0.021975

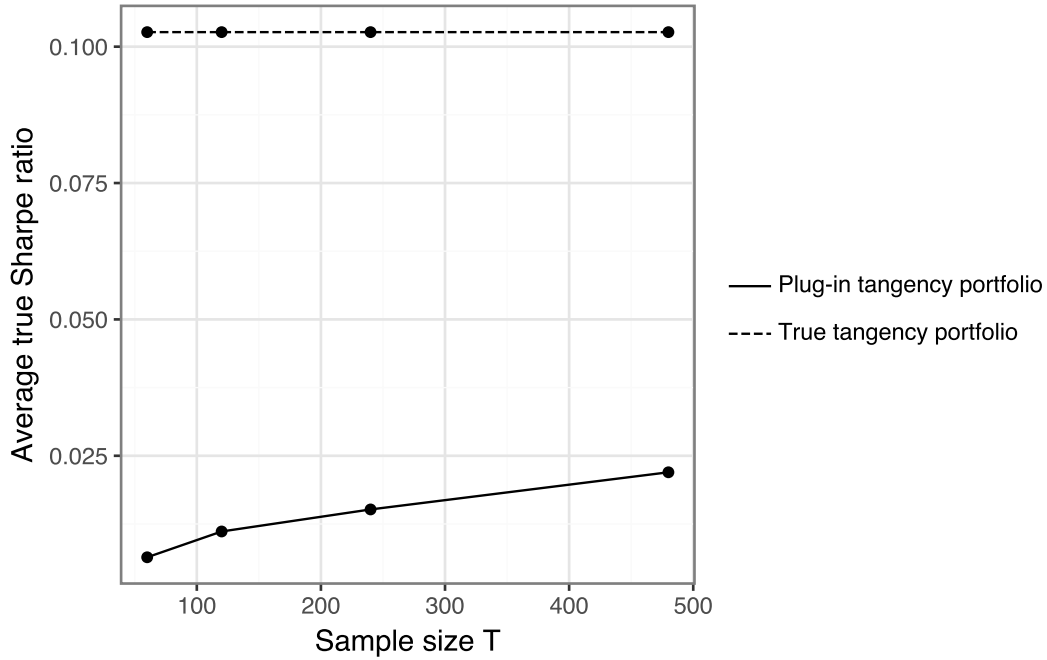


Figure 3: True and plug-in tangency portfolio performance evaluated under the true parameters.

Table 3 reports the average Sharpe ratios of the true tangency portfolio and the plug-in tangency portfolio for different sample sizes.

The results show that the plug-in tangency portfolio performs substantially worse than the true tangency portfolio when T is small. As T increases, the plug-in Sharpe ratio improves because the estimates $\hat{\mu}$ and $\hat{\Sigma}$ become more accurate.

Figure 3 illustrates the same pattern. The performance of the plug-in tangency portfolio increases with the sample size, but remains below the performance of the true tangency portfolio.

Exercise 7

To study the robustness of different portfolio construction methods, we compare the plug-in tangency portfolio with several alternative strategies.

The first benchmark is the equally weighted portfolio, defined as

$$w_{EW} = \frac{1}{N}\mathbf{1}.$$

This portfolio assigns equal weights to all assets and does not depend on estimated parameters.

The second strategy is the global minimum variance portfolio, which minimizes portfolio variance subject to the weights summing to one. The weights are given by

$$w_{GMV} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.$$

This strategy depends only on the covariance matrix and does not require estimates of expected returns.

The third strategy is the inverse-volatility portfolio, where assets with higher volatility receive lower weights. The weights are defined as

$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^N 1/\sigma_j}.$$

Table 4: Average true Sharpe ratios across portfolio strategies and sample sizes.

strategy T	Equally weighted	Global minimum variance	Inverse volatility	Plug-in tangency
60	0.1018	0.0191	0.102	0.0052
120	0.1018	0.0371	0.102	0.0104
240	0.1018	0.0428	0.102	0.0135
480	0.1018	0.0454	0.102	0.0237

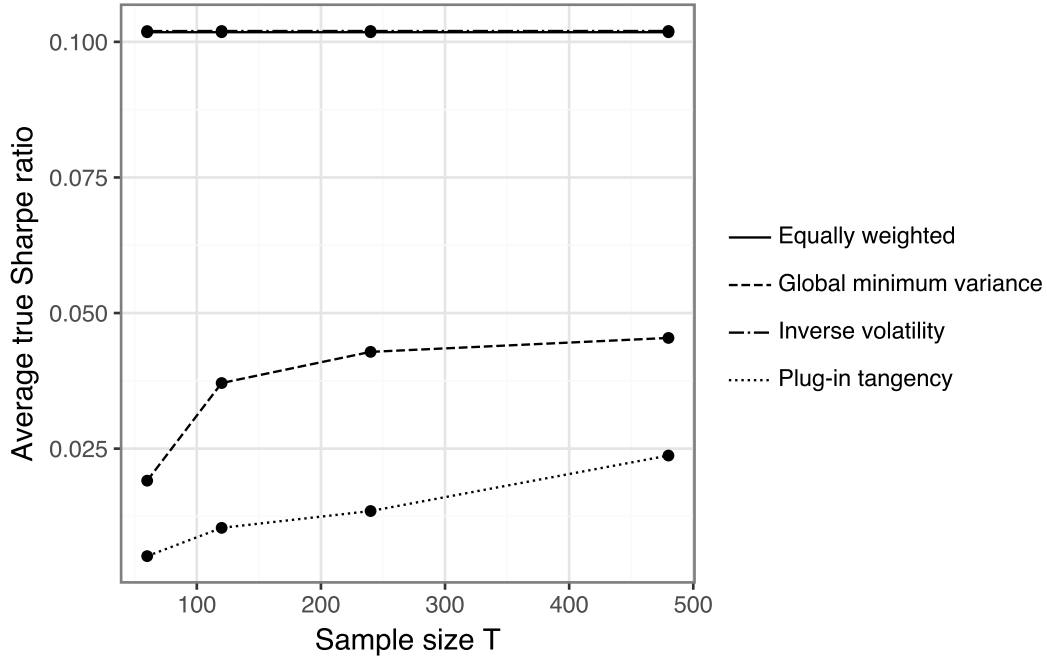


Figure 4: Average true Sharpe ratio across portfolio strategies.

Figure 4 compares the average Sharpe ratios of the different strategies across sample sizes.

The results show that the plug-in tangency portfolio is the most sensitive to estimation error. In contrast, the equally weighted and inverse-volatility portfolios perform more consistently across different sample sizes because they rely less on estimated parameters. As the sample size increases, the performance of the plug-in tangency portfolio improves and gradually approaches the true optimum.